

Test Project

<Mobile Applications Development>

Friendly competition 2024(Guangzhou, China)

Module D

(2.5 hours)

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Introduction

The "MY Health DATA" app aims to record and analyze user health information to support a healthy lifestyle.

The app is currently in the early stages of development.

This initial version aims to provide daily target setting (daily steps, daily water intake), BMI calculation, automatic step tracking, heart rate monitoring, food logging, and water intake record.

As a developer, you have been provided with a wireframe, and your task is to develop the features that will be used in the app based on this wireframe. Additionally, the project manager has asked you to check whether animation effects can be applied on some pages.

Description of project and tasks

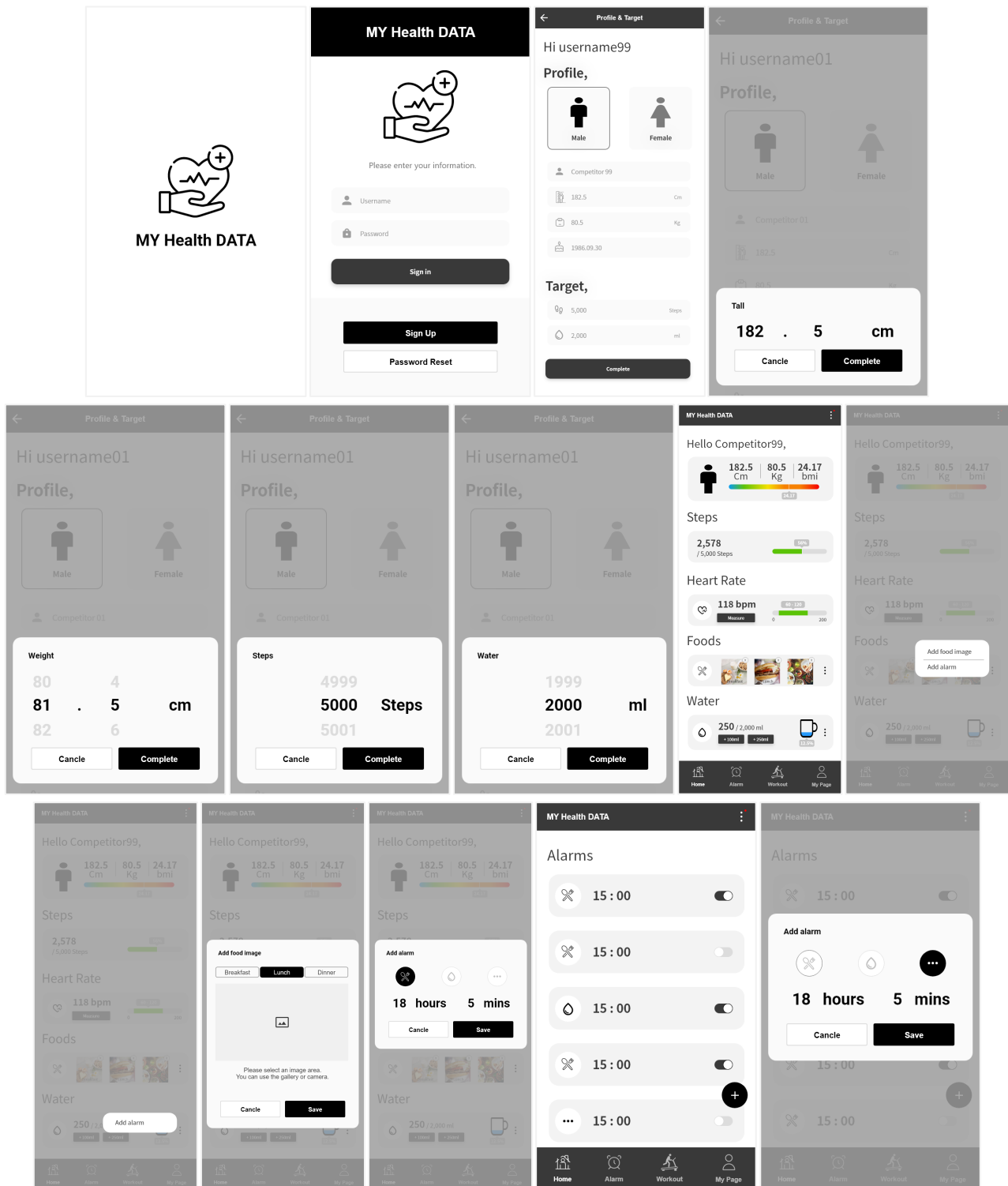
Create all elements referring to the wireframe (no need for 100% match).

The margins, layout, color, font, and font size of the elements do not need to match the wireframe exactly.

Use Bearer Authentication for API authentication.

There is no need to create a sign-up feature or page; you can sign up using the provided API.

If there is no wireframe or missing elements, apply general mobile app development practices and UI/UX standards.



<Wireframe>

1. General Requirements

1. Description

- 1.1 When the app is launched, the splash page appears.
- 1.2 After signing in, move to the profile and target-setting page.
- 1.3 After signing in and closing the app, the sign-in procedure is skipped when the app is launched again (after setting profile and targets).
- 1.4 Develop the Home page, Alarm page, Workout page, and My page.
- 1.5 The top and bottom areas exist on these four pages and remain fixed when scrolling.
- 1.6 The Home page is composed of the top area, profile area, Steps area, Heart Rate area, Foods area, Water area, and navigation area.
- 1.7 The Alarm page consists of the top area, alarm list, bottom area, a floating button, and navigation area.
- 1.8 The Workout page and My page have titles and are empty.
- 1.9 If validation fails or an error occurs, display an appropriate message.

2. Requirements (Home Page, Alarm Page, Workout Page, My Page)

NO	ELEMENTS	REQUIREMENTS
1	Top Area: 1. "MY Health DATA" text 2. "More" icon menu 3. "Red dot" element	Location: Top of the screen 1. Remains fixed when scrolling.
2	Navigation (Bottom Area): 1. "Home" menu 2. "Alarm" menu 3. "Workout" menu 4. "My Page" menu	Location: Bottom of the screen 1. The menu of the current page is highlighted. 2. Remains fixed when scrolling. 3. All menus include an icon and text elements.

2. Splash Page

1. Page Description:

- 1.1 When the app is launched, the "Splash" page is displayed.
- 1.2 The "Splash" page shows appropriate animations.
- 1.3 After the animation ends, it transitions to the "Sign-in" page or "Home" page.

2. Requirements

NO	ELEMENTS	REQUIREMENTS
1	Splash Page: 1. "MY Health DATA" symbol 2. "MY Health DATA" text	1. The "MY Health DATA" symbol animates within 3 seconds. 2. The "MY Health DATA" text animates within 3 seconds. 3. (Each element animates separately.) 4. If signed out, the page transitions to the Sign-in page within 2 seconds after the animation ends. 5. If signed in, the page transitions to the "Home" page within 2 seconds after the animation ends. (After setting targets upon signing in, the sign-in status is maintained.).

3. Sign-in Page

1. Page Description:

- 1.1 It consists of the top area, the Sign-in form area, and the bottom area.
- 1.2 It conducts validation checks on input fields.
- 1.3 It is developed using the Authentication API.

2. Requirements

NO	ELEMENTS	REQUIREMENTS
1	Top Area: 1. "MY Health DATA" text	Location: Top of the screen
2	Sign-in Form Area: 1. "MY Health DATA" symbol 2. "Please enter your information." text 3. "Username" input box 4. "Password" input box 5. "Sign in" button	Location: Center of the screen 1. The "Username" input box has an icon and placeholder. 2. The "Password" input box has an icon and placeholder. 3. Tapping the "Sign in" button initiates validation checks (all input fields must be filled). 4. It is developed through an API. 5. Successful sign-in redirects to the "Profile and Target" page. 6. A failure to sign in displays an appropriate message.
3	Bottom Area: 1. "Sign Up" button 2. "Password Reset" button	Location: Bottom of the screen 1. Tapping the "Sign Up" or "Password Reset" buttons displays a "Coming Soon" message.

4. Profile & Target Page

1. Page Description:

- 1.1 It consists of the profile area, the target area, and the bottom area.
- 1.2 It is developed using the Profile & Target API.

2. Requirements

NO	ELEMENTS	REQUIREMENTS
1	Top Area: 1. "Back" button 2. "Profile & Target" text 3. "Hi {username}" text	1. Clicking the "Back" button returns to the "Sign-in" page. 2. The "Hi {username}" text displays with the user's information included.
2	Profile Area: 1. "Profile" text 2. "Male, Female" selection box 3. "Username" input box 4. "Height" input box 5. "Weight" input box 6. "Birthday" input box Number Input Dialog: 7. "Title, Unit" text 8. "Number" input box (numeric) 9. "Cancel, Complete" buttons	1. The "Male, Female" selection box includes icons and highlights upon selection (only one can be chosen). 2. The "Height" input box includes an icon and units. 3. The "Weight" input box includes an icon and units. 4. The "Birthday" input box includes an icon. 5. Height can be entered using a Number Input Dialog (decimal input allowed). 6. Weight can be entered using a Number Input Dialog (decimal input allowed). 7. The outer area of the Number Input Dialog is dark, and tapping the Cancel button or outside area can cancel the operation. 8. Weight can be adjusted by swiping the numbers in the input.
3	Target Area: 1. "Steps" input box 2. "Water" input box	1. The "Steps" input box includes an icon and units. 2. The "Water" input box includes an icon and units. 3. Step goals can be entered using a Number Input Dialog (no decimals). 4. Water goals can be entered using a Number Input Dialog (no decimals). 5. Steps and weight can be adjusted by swiping the numbers in the input.

		6. The outer area of the Number Input Dialog is dark, and tapping the Cancel button or outside area can cancel the operation.
4	Bottom Area: 1. "Complete" Button	1. Tapping the "Complete" button initiates validation checks (no field can be empty). 2. It saves through the Profile and Target API, and upon completion, moves to the "Home" page. 3. It is developed through an API.

5. Home Page

1. Page Description:

1.1 It consists of the profile area, Steps area, Heart Rate area, Foods area, and Water area.

1.2 It is developed using the Home API, Step API, Heart Rate API, Food API, Water API, and Alarm API.

2. Requirements

NO	ELEMENTS	REQUIREMENTS
1	Profile Area: 1. "Hello {username}," text 2. "Gender" image 3. "Height, Unit" text 4. "Weight, Unit" text 5. "BMI, Unit" text 6. "BMI" graph, "BMI" balloon text	1. "{username}", "Height", and "Weight" are generated via an API. 2. BMI calculations are accurately performed: $\text{BMI} = \text{Weight (kg)} / \text{Height (m)}^2 = \text{Weight (kg)} / (\text{Height (m)} * \text{Height (m)})$ Example: $24.17 = 80.5 / (1.825 * 1.825)$ 3. The "BMI" graph is created with a gradient. 4. The position of the "BMI" graph matches the BMI value. (Graph range: minimum 10, maximum 60)
2	Steps Area: 1. "Current Steps" text 2. "Goal Steps" text 3. "Step Achievement Rate" graph	1. Developed via an API. 2. The achievement rate (%) is calculated and displayed. 3. The calculated achievement rate is reflected in the graph. 4. Counts are calculated with each walk updated every 5 seconds (Generate a random value between 0 and 10).

	4. "Achievement Rate (%)" balloon text	5. Every 5 seconds, as the Step count updates, a counter animation sequentially increases the numbers. 6. Every 5 seconds, as the Step count updates, the graph and achievement rate are updated. 7. Every 5 seconds, the updated count is saved via the API.
3	Heart Rate Area: 1. "Heart Rate" text 2. "Heart Rate" icon 3. "Last bpm" text 4. "Measure" button 5. "Heart Rate" graph 6. "Min-Max" balloon text	1. "Last bpm" and "Min-Max" balloon text are generated via an API. 2. "Heart Rate" graph's data is calculated. 3. Tapping the "Measure" button measures the heart rate using the API, updating the last bpm and the graph. (API generates arbitrary values.)
4	Foods Area: 1. "Foods" text 2. "Foods" icon 3. "Foods" image list 4. "Delete Food Image" button 5. "Food Image" type 6. "More" icon menu Add Food Image Dialog: 7. "Breakfast", "Lunch", "Dinner" selection boxes 8. "Selected Image" area and description text 9. "Cancel", "Save" buttons Add Alarm Dialog: 10. "Foods", "Waters", "Other" selection boxes 11. "Hour", "Minute" input boxes and "hours", "mins" text 12. "Cancel", "Save" buttons	1. Developed through an API. 2. Tapping the delete button removes the image. 3. Images can be swiped if they exceed the area. 4. "More" menu includes "Add Food Image" and "Add Alarm" items. 5. Tapping "Add Food Image" menu item displays the Add Food Image modal dialog. 6. Tapping "Add Alarm" menu item displays the Add Alarm modal dialog. Add Food Image: 7. The dialog can be closed by tapping the outer area or the "Cancel" button. 8. One of "Breakfast", "Lunch", or "Dinner" can be selected. 9. Tapping the image area allows choosing from the gallery. 10. If an image is selected from the gallery, it is displayed in the image area. 11. Tapping the "Save" button saves the image via the API and updates the Foods area. Add Alarm Dialog:

		12.The dialog can be closed by tapping the outer area or the "Cancel" button. 13.The "Foods" selection box is selected. 14.Only one can be chosen from "Foods", "Waters", "Other" selection boxes, which are icon-based. 15.Tapping the "Save" button adds the alarm via the API. 16.Once the alarm is added, a message is displayed.
5	Water Area: "Water" text "Water" icon "Current Water Intake" text "Water Goal Intake" text "Cup" icon, "Achievement Rate" balloon text " + 100ml", "+ 250ml" buttons Add Alarm Dialog: "Foods", "Waters", "Other" selection boxes "Hour", "Minute" input boxes and "hours", "mins" text "Cancel", "Save" buttons	- The Water area is developed via an API. - Tapping the "+ 100ml", "+ 250ml" buttons allows adding to the current water intake, and it's made through the API. - The water intake achievement rate is calculated and shown in blue in the cup. - When the achievement rate changes, an animation of water filling the cup is activated. "More" menu includes the "Add Alarm" item. - Tapping the "Add Alarm" menu item displays the Add Alarm modal dialog. Add Alarm Dialog: - The dialog can be closed by tapping the outer area or the "Cancel" button. - The "Waters" selection box is selected. - Only one can be chosen from "Foods", "Waters", "Other" selection boxes, which are icon-based. - Tapping the "Save" button adds the alarm via the API. - Once the alarm is added, a message is displayed.

6. Alarm Page

1. Page Description:

- 1.1 It consists of the alarm list area.
- 1.2 It is developed using the Alarm API.

2. Requirements

NO	ELEMENTS	REQUIREMENTS
1	Alarm List Area: "Alarm" icon "Alarm Time" text "Active/Inactive" radio button "+" Floating button Add Alarm Dialog: "Foods", "Waters", "Other" selection boxes "Hour", "Minute" input boxes and "hours", "mins" text "Cancel", "Save" buttons	- The list is configured through an API. - The "Active/Inactive" feature operates via the API. - The "+" Floating button scrolls with the screen. - Tapping the "Add alarm" menu item displays the Add Alarm modal dialog. Add Alarm Dialog: - The dialog can be closed by tapping the outer area or the "Cancel" button. - The "Other" selection box is pre-selected. - The "Foods", "Waters", "Other" selection boxes are composed of icons and only one can be selected. - Tapping the "Save" button adds the alarm through the API. - Upon completion of the addition, a message is displayed and the list is updated.

Instructions to the Competitor

- Please refer to what the competition has to offer.